

DIFFERENTIAL DIAGNOSIS

The differential diagnosis of staphylococcal toxic shock syndrome (TSS) includes: - Septic shock - Staphylococcal exfoliative syndromes - Rocky Mountain spotted fever - Viral hemorrhagic shock - Measles - Leptospirosis - Stevens-Johnson syndrome

Although **Kawasaki syndrome** shares several clinical features—such as extremity swelling and desquamation of the palms and soles during convalescence—it differs in key aspects: the fever is more prolonged, and there is an absence of diarrhea and hypotension.

Streptococcal-mediated syndromes, including scarlet fever and especially streptococcal TSS, can closely mimic staphylococcal TSS.

In any patient presenting with fever, rash, and hypotension, a thorough evaluation for potential staphylococcal and streptococcal infection sites is essential. This includes: - Careful examination of surgical wounds, even in the absence of overt signs of infection - Vaginal examination with removal of tampons or other foreign bodies - Vaginal irrigation with saline or povidone-iodine (as recommended in some cases)

TREATMENT AND PROGNOSIS

Management of TSS is primarily supportive and typically requires intensive care. The main goals are hemodynamic support and eradication of *Staphylococcus aureus*.

Historically, high-dose **β -lactamase-resistant anti-staphylococcal antibiotics** (e.g., nafcillin) have been used. However, these agents may increase TSST-1 production *in vitro* due to bacterial lysis. Therefore, **clindamycin**—which inhibits bacterial protein (toxin) synthesis—is often added concomitantly.

Due to rising rates of community-acquired methicillin-resistant *S. aureus* (MRSA), **vancomycin** is frequently recommended as initial therapy. β -

lactamase-resistant penicillins are less effective in high-burden infections compared to clindamycin or vancomycin.

Todd's recommendation: A combination of **vancomycin and clindamycin** should be considered for suspected severe staphylococcal infections pending culture and sensitivity results.

Intravenous immunoglobulin (IVIG) has emerged as a promising adjunctive therapy, likely acting through neutralization of superantigens. It is currently reserved for severe or refractory cases.

Contraindications to IVIG include: - Prior hypersensitivity reactions - Immunoglobulin A (IgA) deficiency

Systemic corticosteroids remain controversial. Their benefit is limited, as superantigen-mediated immune activation is often corticosteroid-resistant.

Prognosis

Staphylococcal TSS is life-threatening but has a mortality rate of approximately **5%**, likely because most affected individuals are otherwise healthy young people. However, **recurrence occurs in up to 20%** of cases.

Preventive measures for women who have had TSS: - Avoid tampon use during menstruation - Avoid diaphragms and contraceptive sponges

Streptococcal TSS

Treatment parallels that of staphylococcal TSS. In cases associated with **necrotizing fasciitis or myositis**, early recognition and **surgical debridement** are critical.

IVIG is increasingly recognized as a vital component of therapy for streptococcal TSS, particularly given its mortality rate, which can exceed **30%**.

Recalcitrant Erythematous Desquamating Disorder (REDD)

First described in 1992, REDD is a toxin-mediated condition with overlapping features of TSS.

Clinical Features:

- Fever
- Hypotension
- Diffuse macular erythema with delayed desquamation
- Ocular and oral mucosal injection
- Strawberry tongue

Microbiology:

Staphylococcus aureus producing **TSST-1**, **staphylococcal enterotoxin A (SEA)**, or **SEB** has been isolated from: - Nasal sinuses - Soft tissues - Blood

Key Differences from TSS:

- Prolonged course (weeks to months)
- Only partial fulfillment of TSS diagnostic criteria
- Often occurs in immunocompromised individuals (many reported cases in patients with AIDS), though cases in immunocompetent individuals exist

Diagnosis:

Established by identifying occult colonization or infection in susceptible hosts through careful clinical evaluation.